

Nicolas Jao

4th Year Automation & Controls Engineering

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PROFESSIONAL SUMMARY

Final-year Automation Systems Engineering Technology student with hands-on experience in PLCs, industrial robotics, and electrical design for manufacturing equipment. Seeking a summer 2026 co-op (May to August) where I can contribute immediately to plant operations and continuous improvement. Expected to graduate December 2026.

EDUCATION

Bachelor of Technology, Automation Systems Engineering Technology Co-op Program

Sep 2022 — Dec 2026

McMaster University & Mohawk College, Hamilton, Ontario

- Concurrent Chemical Engineering Technology Advanced Diploma and ACBSP Accredited Certificate of Business
- FANUC Certified Robot Operator – Level 1
- Specialties: PLCs, Industrial Robotics, Systems Design, Chemical Engineering Technology, IoT Networks

EXPERIENCE

Mecsmart Systems – Electrical Design Co-op

Jan 2025 — Aug 2025

Newmarket, Ontario

- Designed electrical systems for 10+ industrial manufacturing machines such as MIG welders, laser welders, riveters, and end-of-line testers for the automotive industry, working closely with and learning from a licensed P.Eng.
- Drafted 3+ electrical schematics and 5+ panel layouts for the electricians in the workshop using **EPLAN Electric P8**
- Used **SolidWorks** to view and decide where to place sensors and Ethernet I/O blocks on our machines
- Estimated the total current load of 3 projects through power calculations using **Excel** and manual calculations
- Led the design of 3 full electrical schematics of 3 stations of a manufacturing line
- Improved electrical department efficiency by 30%, creating new standard procedures, implementing new **EPLAN** and **Excel** features and updating documentation for future employees
- Gained valuable experience reading 50+ data sheets, wiring diagrams, terminal pinouts, instruction manuals, user manuals, customer quotes and project management technical documents to complete daily tasks
- Collaborated with electricians and the mechanical department to solve daily issues in the build process of projects

BTech Smart Systems Stream – Labs

Sep 2023 — Dec 2026

McMaster University & Mohawk College, Hamilton, Ontario

- Worked from a systems integrator perspective to troubleshoot automated lab equipment, performing failure analysis and preventative maintenance to support equipment uptime
- Calibrated RTDs, thermocouples, and transmitters with **DeviceCare SFE100**, pressure gauges with deadweight testers
- Conducted motor control tests with **RSLogix ladder logic** and wired and configured the Allen-Bradley PLC I/O modules
- Created chemical plant process flow diagrams with **UniSim Design**, implementing reactors, exchangers, and separators
- Tested and tuned PID control on VFD-driven motors to ensure stable and reliable speed control for industrial equipment
- Programmed advanced routines for FANUC robots with the teach pendant and **RoboGuide**
- Implemented industrial IoT lab projects using **Arduino**, **LabVIEW**, and cloud dashboards for sensor monitoring
- Designed logic circuits on FPGAs using **Quartus II** to practice building components of digital electronics
- Assembled and tested embedded systems with **e2 Studio**, performing signal interfacing and real-time control
- Produced technical documentation and reports using **Microsoft Word** and created lab data graphs using **Excel**

PROJECTS

Sushi Assembly Line – CAPSTONE Project

Jan 2026 — Dec 2026

McMaster University, Hamilton, Ontario

- Designing a fully automated sushi production machine integrating material handling, process sequencing, rolling, and slicing with minimal operator input, with project objectives in safety, consistency, and food grade materials

SKILLS

Programming: Allen-Bradley RSLogix 5000, FANUC RoboGuide, e2 Studio, C/C++, Python, Java

Design: EPLAN Electric P8, SolidWorks, AutoCAD, Honeywell UniSim Design, NI LabVIEW, Intel Quartus II

Miscellaneous: FESTO DeviceCare SFE100, Excel, Microsoft Office, MATLAB, P&IDs